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Progress of the bolometer diagnostic system on EAST

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Abstract

Measurements of the total radiated power and its spatial distribution are crucial for fusion research. On the experimental advanced superconducting tokamak (EAST), both the metal foil resistive bolometer and the absolute extreme ultraviolet (AXUV) photodiodes have been used to quantify the radiated power. This article introduces the latest improvement of the bolometer diagnostic system on EAST. It also details the successful design and installation of new divertor AXUV cameras, which are dedicated to the investigation of divertor physics. The shielding components of the bolometer detector have been refined, and the article provides a detailed exposition of the double shielding structures that have been verified as effective in microwave shielding. Additionally, the changes in the radiated power distribution in the divertor region during the plasma detachment process are measured using the divertor AXUV camera. Finally, the radiated power measured by the AXUV detector and metal foil resistive bolometer are compared, and different detector performances are presented.

Keywords: radiated power, bolometer diagnostic, microwave shielding, plasma detachment, sensitivity degradation

(Some figures may appear in colour only in the online journal)

1. Introduction

Plasma radiation is one of the main energy loss ways in experimental fusion devices. It is necessary to determine the total radiated power as well as the radiation emission profile for studying the power balance, impurity transport, radiation instabilities, and other key physics processes [1, 2]. The metal foil resistive bolometer and the AXUV detector are extensively used for radiated power measurements in present fusion devices, such as DIII-D [3], LHD [4], ASDEX-Upgrade (AUG) [5], and others. Both detector types are deployed in the bolometer diagnostic systems on EAST [6].

Conventional metal foil resistive bolometers consist of a radiation-absorbing foil (typically gold or platinum) deposited onto a thin substrate foil, such as kapton or mica, featuring resistance meanders on its reverse side. The radiation-hard metal foil bolometer detector designed for ITER device, incorporating novel materials like SiN membranes, has seen substantial advancements over the past decade [7]. In the case of EAST, a 4 μm platinum (Pt) foil is selected as the detector absorber, with Si_3N_4 serving as the substrate material, produced by German IPT GmbH. It has a fairly linear response to photons spanning from infrared to soft X-ray with a temporal resolution of several milliseconds (~ ms) [8, 9]. The 4 μm Pt foil effectively can absorb nearly all photons below 5 keV, maintaining an absorption ratio of

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approximately 63% for 10 keV photons. The AXUV photodiodes are based on silicon p-n junctions. The incident photons can generate electron-hole pairs in the detector's active region, resulting in a photocurrent output proportional to the incident radiated power. The AXUV detectors exhibit a flat spectral sensitivity across a photon energy range of 200 eV to 6 keV. Below 200 eV, the sensitivity diminishes and varies significantly with photon energy, while above 6 keV, the sensitivity rapidly declines due to the photon absorption length exceeding the effective silicon thickness [5]. The AXUV detectors offer a rapid response time of ~ 0.5 μs. Their compact structure and high time resolution suit them for studying rapid processes, such as radiation instabilities and disruption events [10].

During the EAST lower divertor upgrade in 2021 [11], both the AXUV and resistive bolometers were upgraded to fulfill the demands of future experiments. Figure 1 illustrates a top-down view of the EAST toroidal windows, along with the toroidal installation positions of the AXUV and bolometer diagnostics. The figure also shows the locations of the auxiliary heating systems, including ion cyclotron resonance heating (ICRH), electron cyclotron resonance heating (ECRH), lower hybrid wave heating (LHW), and neutral beam injection (NBI). The original EAST AXUV diagnostic system comprised horizontal cameras (P-port) and upper vertical cameras (upper C-port). As part of the upgrade, the new toroidal cameras (J-port) and additional lower divertor cameras were installed. The new toroidal AXUV camera, situated near the scattered pellets injection (SPI) system, is designed for future disruption mitigation research and shares a similar field of view and structural compositions as the horizontal P-port cameras [2]. The new lower divertor AXUV cameras are specifically designed to obtain the radiation distribution within the divertor region, where radiation constitutes a significant fraction of heat dissipation in discharges with injected impurities. The detailed design of the divertor cameras and corresponding measurement results will be described in the following sections. Regarding the bolometer diagnostic system, enhancements were primarily directed toward optimizing the support of the detector and shielding structures to prevent microwave interference.

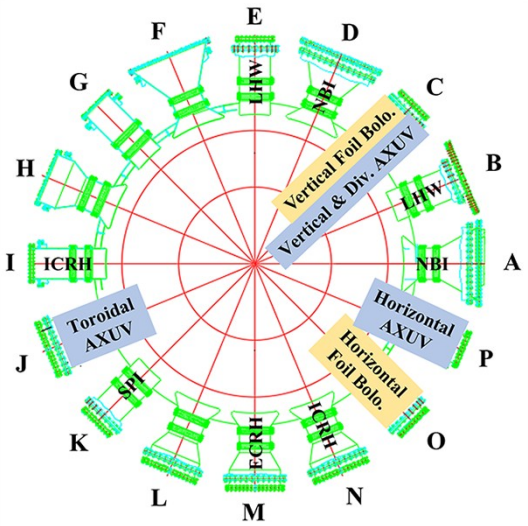


Figure 1. Top-down view of the EAST toroidal windows, indicating the installation positions of the AXUV diagnostic (in blue) and foil bolometers (in yellow).

This article is structured as follows. Section 2 details the design of the lower divertor AXUV cameras. Section 3 shows the optimized bolometer shielding structure designed for microwave shielding. Section 4 gives a comparison of measurement results from the two distinct detector systems. The article concludes with a summary and conclusion in the final section.

2. Design of the lower divertor AXUV camera

For the passive radiation measurement diagnostic system in tokamaks, detectors receive plasma radiation along the line-of-sight (LOS) through an aperture, with a finite viewing angle [12]. To ensure reliable measurements of the total radiated power and accurate tomographic reconstruction of the radiation profile, the LOS needs to cover the plasma poloidal cross-section from multiple viewing angles. The EAST AXUV diagnostic system includes horizontal port cameras, upper vertical port cameras, and new lower divertor cameras, with their LOS layout depicted in figure 2(a). The horizontal cameras, installed in the P-port, utilize four AXUV16ELG detectors, totaling 64 channels. The vertical arrays, installed in the upper C-port, consist of two cameras equipped with AXUV20ELG detectors, summing up to 40 channels. The spatial resolution δ_r for both the P-port and C-port cameras is about 3–4 cm. In this context, δ_r represents the distance between adjacent LOSs in the radial position at R = 1.85 m (R is the major radius of EAST) for the horizontal cameras, and at Z = 0 m for the vertical cameras. As illustrated in figure 2(a), three new lower divertor cameras are designed to cover the inner target region, outer target region, and the X-point region, respectively. These cameras are installed near the lower C-port, occupying the same toroidal position as the vertical AXUV cameras. The spatial resolution δ_r for the divertor cameras is about 1–2 cm, where δ_r is defined as the

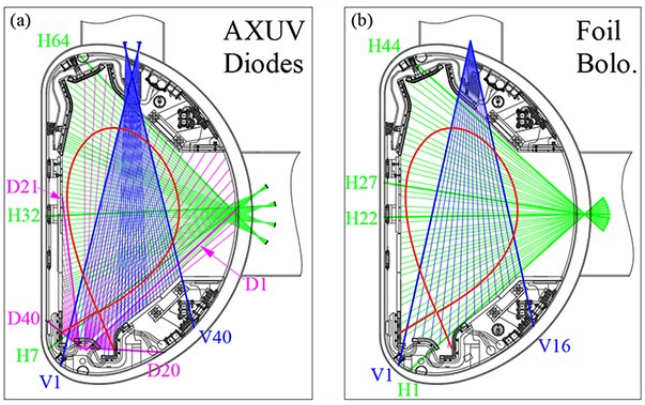


Figure 2. The EAST poloidal cross-section, including the LOS for the AXUV diagnostic (a) and metal foil bolometer diagnostic (b). The red line indicates the last closed flux surface (LCFS) of a typical lower single-null (LSN) plasma configuration.

distance between two neighboring LOSs along the X-leg line.

Given the constrained space beneath the divertor, lower divertor cameras use compact AXUV20ELG detectors. Figure 3 shows the detector housing structure, wherein the detectors are securely mounted within compact stainless steel shielding housings. These detectors receive plasma radiation through a narrow aperture with a size of 1 mm × 5 mm, situated 20 mm away from the detector surface. The detector housing is fixed on the support structure of the lower divertor cassettes through brackets, and it is placed under the dome region to shield the detectors from direct radiation and prevent plasma dust accumulation. Here, a specialized divertor cassette support frame is designed to provide the necessary installation space. The LOS makes full use of the gap between the two divertor cassettes and part of the dome tiles are trimmed to square openings for the LOS pathway. The square openings in the dome allow the LOSs to view both the inner and outer targets of the divertor, as well as the X-point area. Furthermore, an active water-cooling structure is designed around the detector supports, passing through the inside of the detector housing from one side and then returning from the other, to ensure a safe operating temperature below 80 °C during the high-temperature first wall baking process (150 °C–200 °C). The detector system and its associated devices are insulated, and the signal cables are transmitted from the lower port to the outside of the vacuum vessel. Figure 4 shows the AXUV detector housing photos before and after installation.

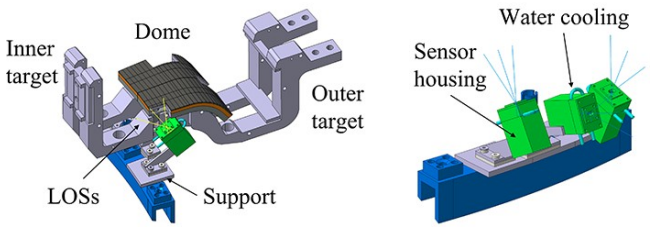


Figure 3. The EAST lower divertor cassettes (left) and the AXUV detector shielding housing box (right).

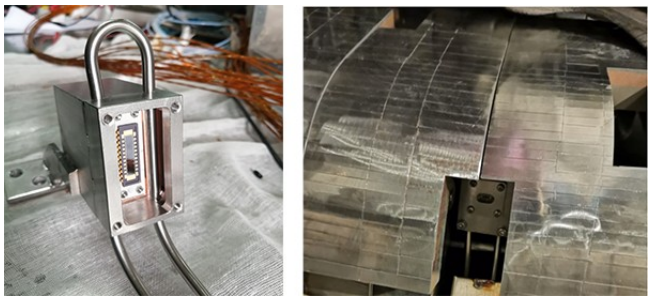


Figure 4. The detector housing photos before (left) and after installation (right).

3. Optimizations of the bolometer detector shielding

The foil bolometer diagnostic system on EAST consists of a 44-channel horizontal array, installed in the O-port to fully cover the plasma poloidal cross-section, as well as a 16-channel array integrated into the upper vertical C-port. Figure 2(b) shows the LOS distribution for the EAST metal foil bolometer, with a spatial resolution δ_r of about 3–6 cm. The foil resistive bolometer detectors, positioned within the vacuum vessel, are sensitive not only to the thermal deposition of photons but also to the interference of microwaves [13]. This interference can lead to detector heating due to eddy currents generated by the interaction between the absorber and microwaves in several skin depths. Experimental observations confirm that microwave interference on resistive bolometers is unavoidable. Three types of microwave bands are applied for plasma heating and current drive on EAST, including 28 MHz ICRH, 140 GHz ECRH, and 2.45 GHz/4.6 GHz LHW. Unabsorbed microwaves will be scattered in the vacuum vessel and can enter into the detectors through the front collimator apertures or some back slits, thereby affecting the accuracy of measurements. Consequently, the design of shielding structures is critical for the reliable operation of resistive foil bolometer detectors.

Figure 5 shows the optimized shielding structure for the foil bolometers. The detectors are mounted on a single arc-shaped copper support plate, with two groups of water-cooling pipes integrated into two sides of the support plate to mitigate the heat accumulated during long pulse discharges. The measured cooling effect is given in section 4.2. To reduce the impact of microwaves on detectors after multiple scattering, the sensors are encased in an inner shielding box and an outer shielding box. Additionally, the inner surface of the shielding box is coated with a 20 μm thick layer of TiO_2, which further absorbs the microwave and reduces its reflection [14]. Both the inner shielding box's collimating aperture (3 mm × 10 mm) and the larger aperture of the outer shielding box are equipped with copper mesh grid layers. The mesh grid can reflect the microwaves and also can attenuate the intensity of incident photons. The mesh grid opening size g and the mesh wire diameter d are two important design parameters. The microwave shielding efficiency is influenced by the ratios g/λ and g/d. Typically, for effective shielding, g should be less than half the wavelength λ of the microwave ($g < \lambda/2$) [15]. The photon transmittance T is calculated by the formula $T = (1-d/g)^2\cos(\theta)$, where θ is the incident angle. A higher d/g ratio results in lower photon transmittance but improves the shielding effect of the metal

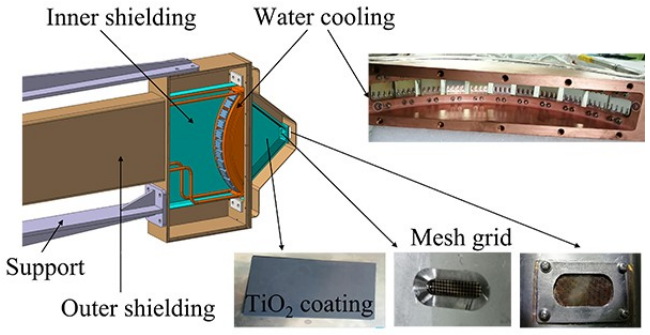


Figure 5. The shielding structure of foil bolometers.

grid against microwaves. Therefore, a balance must be achieved between the shielding efficiency of microwaves and the transmission of photons. For this reason, the mesh grids with $g/d = 2.96$ and $g/d = 7.35$ are chosen for the outer and inner aperture, respectively. The outer mesh grid dimensions are $g = 475$ μm and $d = 160$ μm, while the inner mesh grid dimensions are $g = 559$ μm and $d = 76$ μm. The optical transmission fraction of the mesh grids is calibrated using visible LED light under normal incidence conditions. The results are presented in table 1, showing good agreement between the measured transmittance and the calculated value for visible light.

Figure 6 illustrates a comparison of the bolometer signal performance before and after microwave shielding optimization during two EAST discharges with LHW injection. Discharge #89502 represents the condition before bolometer shielding optimization, while discharge #123090 represents the improved bolometer double shielding condition. As shown in figure 6(b), the bolometer signal distortion occurs after LHW injection with previous shielding structures. Normally, the radiation level increases after LHW injection, as indicated by the AXUV signal in figure 6(c). However, the distortion bolometer signal drops sharply to negative values. Bolometer signal H27 and AXUV signal H37, as well as bolometer H22 and AXUV H32, view nearly the same plasma region, as shown in figure 2. With the enhanced shielding, the bolometer signals return to normal after LHW injection, and the trend changes in the bolometer signal align consistently with those of the AXUV signals. The two signal peaks at approximately 1.8 s and 3.2 s in figures 6(e) and (f) are attributed to a sudden impurity burst following LHW injection. These demonstrate that the reinforced double shielding structure is effective for microwave shielding.

Table 1. The measured photons transmittance of mesh grid.

Mesh grid LPI	Transmittance	
	Visible light	Calculated
40 ($g/d = 2.96$)	0.58	0.56
40 ($g/d = 7.35$)	0.81	0.78

4. Experiment results

4.1. The measurement of radiation distribution in the lower divertor region

Two of the newly installed divertor AXUV cameras, designed to observe the inner and outer regions of the divertor, have successfully collected valid data during the recent EAST experiments. The divertor AXUV detector signal noise (±60 mV) is higher than that of the horizontal AXUV detector (±20 mV) at the same amplifier gain, primarily due to the challenging environmental conditions and the inadequacy of the cable shield. Figure 7 shows the evolution of measured parameters for EAST discharge #101807, which is characterized by a lower single-null (LSN) configuration with neon (Ne) gas puffing [16] to induce plasma detachment. The plasma parameters include plasma current I_p = 400 kA, toroidal magnetic B_t = 2.4 T, and line-averaged electron density $n_e = 4\times10^{19}$ m^{-3}. The injected power of NBI is approximately 2.8 MW. The signals displayed in figure 7, from top to bottom, represent the plasma current I_p and line-averaged electron density n_e, the radiation intensity in bulk

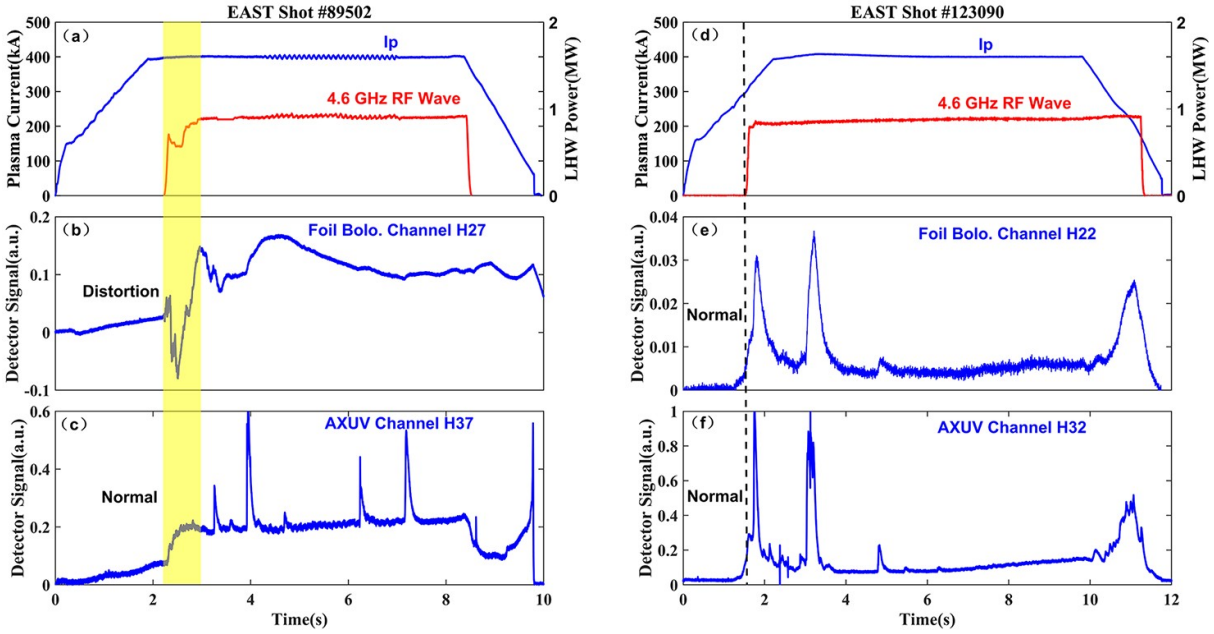


Figure 6. Comparison of metal foil bolometer detectors signals before (left) and after (right) microwaves shielding optimization. (a) and (d) Plasma current (I_p) (blue) and LHW power (red). (b) and (c) Metal foil bolometer detector signals of channel H27 and AXUV detector signals of channel H37 of discharge #89502. (e) and (f) Metal foil bolometer detector signals of channel H22 and AXUV detector signals of channel H32 of discharge #123090.

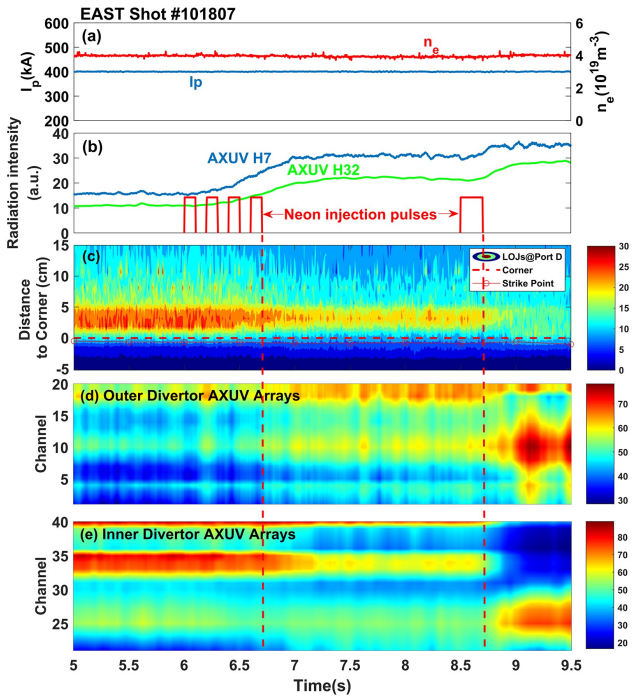


Figure 7. The evolution of plasma parameters for EAST shot #101807. (a) Plasma current I_p (blue) and line-averaged electron density n_e (red). (b) The radiation intensity from AXUV H7 (blue) and AXUV H32 (green) and Ne injection pulses(red). (c) Saturated ion current density (J_s) distribution. The position of the divertor target plate corner and the strike point position are marked with a red horizontal dashed line and a circle line respectively. (d) and (e) The evolution of the radiation intensity distribution in the divertor region.

plasma measured by AXUV H7 (passing through X-points) and H32 (plasma core region) and Ne injection pulses, the saturated ion current density (J_s) measured by divertor Langmuir probes, and the evolution of the radiation intensity distribution in the divertor region. The vertical dotted lines indicated the end time of two Ne injection pulses. In this discharge, it is evident that the I_p and n_e remain stable from 5 s to 9.5 s. The Ne gas is injected in two stages, with the initial injection position near the lower inner divertor target. The first stage of gas injection is divided into four pulse injections, each lasting 0.1 s and separated by 0.1 s interval. The second Ne puffing starts at 8.5 s and lasts 0.2 s. Figure 7 (b) reveals that the radiation intensity in the bulk plasma increases following the Ne gas injection. At the same time, the saturated ion current density shown in figure 7(c) decreases significantly until the plasma reaches a complete detachment state at 8.8 s [17]. Figures 7(d) and (e) show the time-space evolution of radiation intensity in the divertor region measured by the divertor AXUV camera. The localized intense radiation regions in both the inner divertor and outer divertor are located near the strike points before the gas injection. Following the first Ne gas injection at 6.5 s, the radiation intensity in the inner divertor region begins to decrease, while that in the outer divertor region begins to increase. With the plasma achieving complete detachment at 8.8 s, the strong radiation areas near the inner and outer target plates shift upwards and toward the X-point region.

Two-dimension radiation profiles in the divertor region can be reconstructed by tomography. A variety of tomography methods have been devised for line-integrated measurement diagnostics, such as soft X-ray and bolometer diagnostics. These methods include the Maximum Entropy method [18], Minimum Fisher Information [18], Relative Gradient Smoothing (RGS) method [19], Phillips-Tikhonov method [20], and so on. Inverse tomography is an ill-posed problem, with the accuracy of the results highly dependent on the coverage of the LOS. For the EAST AXUV and foil bolometer system, the LOS coverage is insufficient to obtain a two-dimensional radiation profile by using the common pixel method. Consequently, the radiation profile tomography in the divertor region is separated from the main plasma. In the main plasma region, the Bessel-Fourier expansion method is employed, while the finite-elements method is used for the divertor region. The lower divertor region's poloidal cross-section is divided into 20×40 pixels. By integrating the AXUV horizontal and vertical arrays, the radiation distribution in the divertor region is reconstructed by tomography, assuming toroidal radiation symmetry due to the arrays' disparate toroidal positions. Figure 8 shows the radiation distribution at different times for discharge #101807. It also demonstrates that the radiation intensity near the X-point region increases after gas injection, and the radiation emissivity in the bulk plasma edge region above the X-point region increases significantly after detachments.

4.2. Measurement comparison of AXUV and foil bolometer

Since both the AXUV and foil bolometer are utilized on EAST tokamak for the absolute measurement of radiated power, their results are compared for different discharge conditions. Experimentally, it is observed that the total radiated power in the main plasma derived from the AXUV measurement is consistently lower than that obtained from the foil bolometer measurement. This discrepancy arises from the fact that the foil bolometer's energy response range is broader than that of the AXUV detector. Theoretically, the AXUV measurement aligns with the foil bolometer measurement only within the energy range of 200 eV to 6 keV.

Figure 9 shows the temporal evolution of total radiated power in the main plasma for EAST discharge #98958, measured by the AXUV and foil bolometer system. This discharge is a long pulse operation lasting 100 s. The total radiated power in the main plasma is calculated directly from the chord-integrated measurement of the horizontal cameras using the weighted summation method [2], and measurements in the divertor region are excluded. The evolution trend of total radiation aligns closely throughout the entire discharge period. Figure 10 presents the total radiated power evolution in the main plasma for another long-pulse discharge #122296. The signals change trends become different after 200 s. Here, the increasing trend of the foil bolometer after 200 s is not accurate, which is probably caused by the signal background drift due to the temperature increase of the sensor holders. According to the measure-

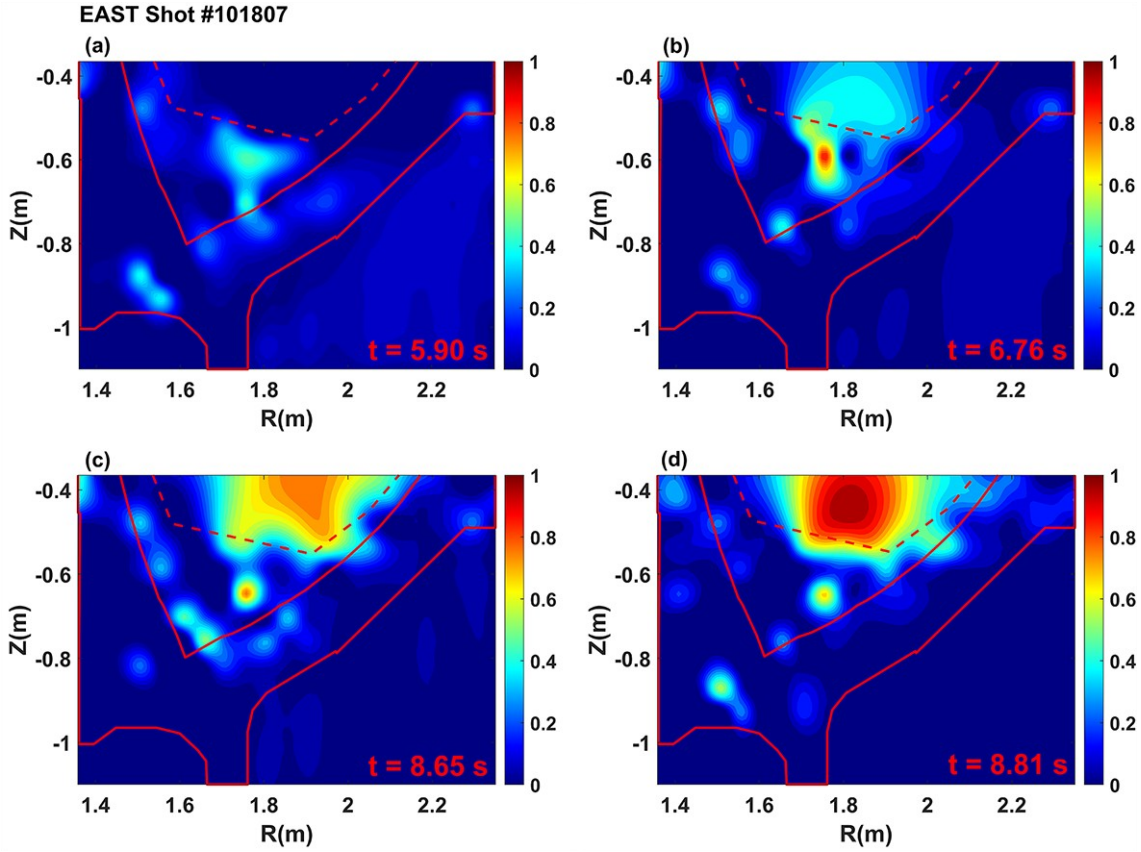


Figure 8. Comparison of the radiation emissivity profiles at different times.

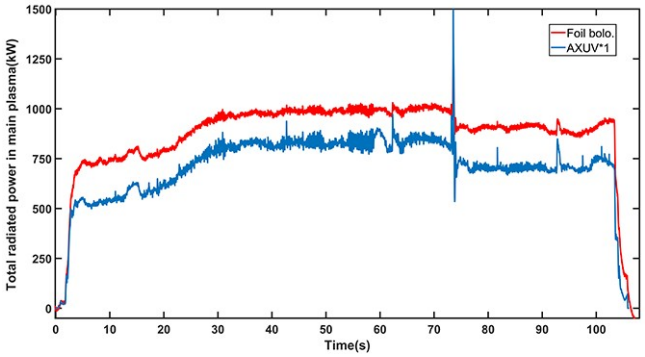


Figure 9. Total radiated power in the main plasma from AXUV and the foil bolometer for discharge #98958.

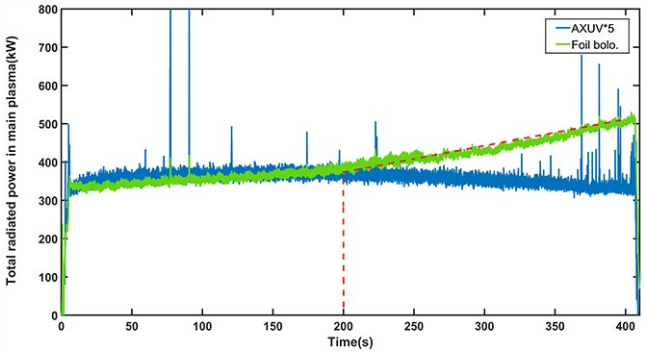


Figure 10. Total radiated power in the main plasma from AXUV and the foil bolometer for discharge #122296.

ment data by thermocouples, the temperature increase of the detector supports is about 1 °C during 100 s long-pulse discharges and about 3 °C–4 °C during 400 s long-pulse discharges without water cooling. The active water cooling system has been found effective in reducing the temperature rise of the detector supports during plasma discharges, ensuring that the detector supports' temperature remains below 40 °C during long-term wall baking with temperatures of 150 °C–200 °C. However, it cannot completely eliminate thermal drift.

The performance of the detector may deteriorate due to the accumulated plasma irradiation effect. The performance change of AXUV detectors is analyzed by cross-calibration with foil resistive bolometer. The sensitivity and cooling time constant (τ) of foil bolometer detectors are shown in figure 11. Detectors selected have worked in EAST for nearly 10 years, and have been exposed to nearly 76000 plasma discharges. It has been found that the parameters of the foil bolometer sensors have changed only slightly (~ 10%), and their performances remain stable. The ratio of the total radiated power measured by the foil bolometer to that measured by the AXUV is compared, and the results are displayed in figure 12. The data below is collected from EAST discharges in the years 2021–2023. During the 2021 experiment campaign, all the AXUV cameras utilized new detectors. From figure 12, it is observed that the measured results of the foil bolometer are higher than the AXUV results by a factor of 2–6. Especially, the difference between the two systems is larger for the EAST high electron temperature operation region (T_e > 5 keV) with ECRH dominated

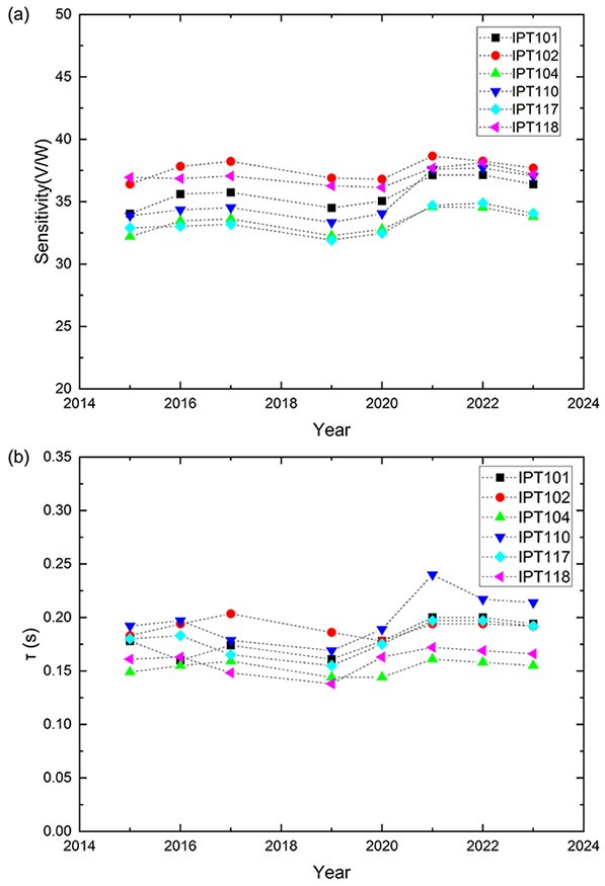


Figure 11. The sensitivity (a) and cooling time constant (τ) (b) of bolometer detectors.

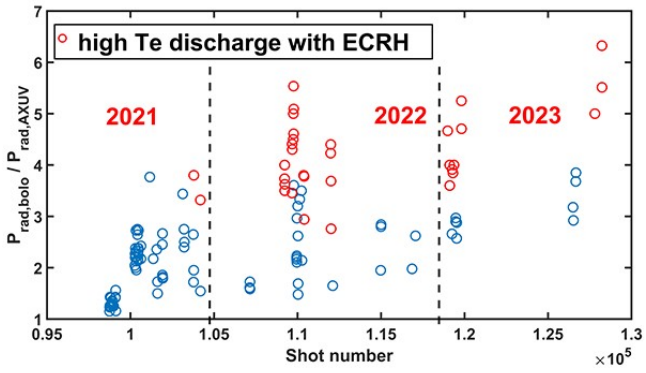


Figure 12. The ratio of the total radiated power in the main plasma of the foil bolometer to the AXUV. P_{rad} means the total radiated power in the main plasma.

core heating. This is caused by the AXUV's response upper limit of 6 keV. The statistical data also indicates that the ratio of the $P_{rad,bolo}$ to the $P_{rad,AXUV}$ has an increasing trend over time, which is predominantly caused by the degradation of the AXUV detector sensitivity. The AXUV detector is not as robust as the foil bolometer, so its response may be affected by radiation bombardment. The AXUV detector response affected by radiation bombardment is examined on AUG [5]. It was found that the sensitivity of the AXUV detector on AUG dropped sharply in the first 50 s of plasma discharges and remained almost constant after ~ 300 s. However, the decrease in AXUV detector sensitivity on EAST is not as severe as it is on AUG. This may be attributed to the lower radiation power in EAST compared to AUG.

5. Summary

The EAST bolometer is upgraded for better performance during EAST lower divertor construction periods. Additional AXUV cameras have been designed and successfully installed on EAST. The new toroidal AXUV camera situated near the SPI injection port enables the study of radiation asymmetry for future disruption mitigation experiments. The divertor AXUV cameras installed to observe both the inner and outer target regions adopt a compact design to accommodate the spatial constraints beneath the divertor cassette. The lower divertor cameras can provide local radiation distribution measurement in the inner/outer divertor regions. When detachment behavior occurs, it is observed that the strong radiation region near strike points moves upwards towards the X-point region. Regrettably, the designed lower divertor AXUV camera to cover the X-point region has not been installed currently due to space conflict. Detachment operation is an essential and effective way to mitigate the high heat flow in high-power injection plasma discharges. The lower divertor AXUV cameras still need further optimization, such as improving the signal-to-noise ratio and increasing the LOS coverage. The radiation pattern in the divertor region and the stable X-point radiation mechanism will be further researched in the future.

The shielding structures of the bolometer detector have been optimized to reduce the entry of microwaves into the detector. The adoptions of double shielding housing, microwave absorbing material coating, and the double mesh grid in front of the apertures have proven to be necessary and effective measures in reducing microwave interference. The experiment data demonstrates that the shielding design improvements for the bolometer system are necessary and effective in weakening microwave interference. At present, the LOS coverage of the foil resistive bolometer is limited, making tomographic reconstruction challenging. The consideration of adding more edge coverage with vertical cameras will be a priority in the upcoming EAST upper divertor upgrades. In long-pulse plasma operations, the background thermal drift of the foil bolometer detector signal cannot be entirely suppressed by active water-cooling, which presents another challenge for the future fusion reactor bolometer design.

Acknowledgments

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